
Seismic Response Spectrum Analysis Report

Ground Acceleration Record, Damping $\zeta = 5\%$

Samples	12,438
Sampling step	$\Delta t = 0.005\text{ s}$
Total duration	$T_d = 62.19\text{ s}$
Peak ground acceleration	$\text{PGA} = 0.713\text{ g} \approx 6.996\text{ m/s}^2$
Damping ratio	$\zeta = 5.0\%$
Solver	Nigam–Jennings exact piecewise-linear SDOF
Period range	$0.02\text{ s} \leq T \leq 5.0\text{ s}$ (200 log-spaced)

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1. Introduction

This report documents the seismic response spectrum analysis carried out on the supplied ground acceleration time history. The objective is to characterise the frequency content and damage potential of the record by transforming the raw time-domain signal into four engineering-relevant quantities used in earthquake-resistant design and assessment:

1. the **ground acceleration time history** itself,
2. the **spectral displacement** $S_d(T)$,
3. the **spectral velocity** $S_v(T)$ and its pseudo counterpart $PS_v(T)$,
4. the **spectral acceleration** $S_a(T)$ (absolute) and its pseudo counterpart $PS_a(T)$.

All response spectra are computed for a viscous damping ratio of $\zeta = 5\%$, the conventional reference value for reinforced-concrete and steel structures in modern codes such as Eurocode 8, ASCE 7 and IS 1893.

1.1 Scope and limitations

The analysis is purely a *linear* elastic single-degree-of-freedom (SDOF) response. Inelastic effects, soil-structure interaction and duration-related damage indices are outside its scope. The report provides the spectra in tabular form (`response_spectra.csv`) so that they can be directly used in modal-response-spectrum analysis of multi-degree-of-freedom buildings.

2. Input ground motion

2.1 Record properties

The record consists of 12,438 acceleration samples written one per line. The sampling interval was taken as $\Delta t = 0.005$ s (200 Hz); the resulting duration is

$$T_d = (N - 1) \Delta t = 12,437 \times 0.005 = 62.185 \text{ s}.$$

The values were interpreted in units of g ; conversion to SI was performed with $g = 9.81 \text{ m/s}^2$. Basic statistics are summarised in Table 1.

Table 1. Summary of the recorded ground motion.

Quantity	Value	Units
Number of samples, N	12,438	–
Sampling interval, Δt	0.005	s
Total duration, T_d	62.19	s
Peak ground acceleration, PGA	0.713	g
PGA (SI)	6.996	m/s^2

2.2 Time-history plot

Figure 1 shows the ground acceleration as a function of time. A strong-motion phase is concentrated within the first ~ 30 s of the record, with several pulses approaching the peak value. A small DC offset and any low-frequency drift have been ignored at this stage; for the spectra below it is sufficient that the mean of the record is close to zero, which a manual inspection confirms.

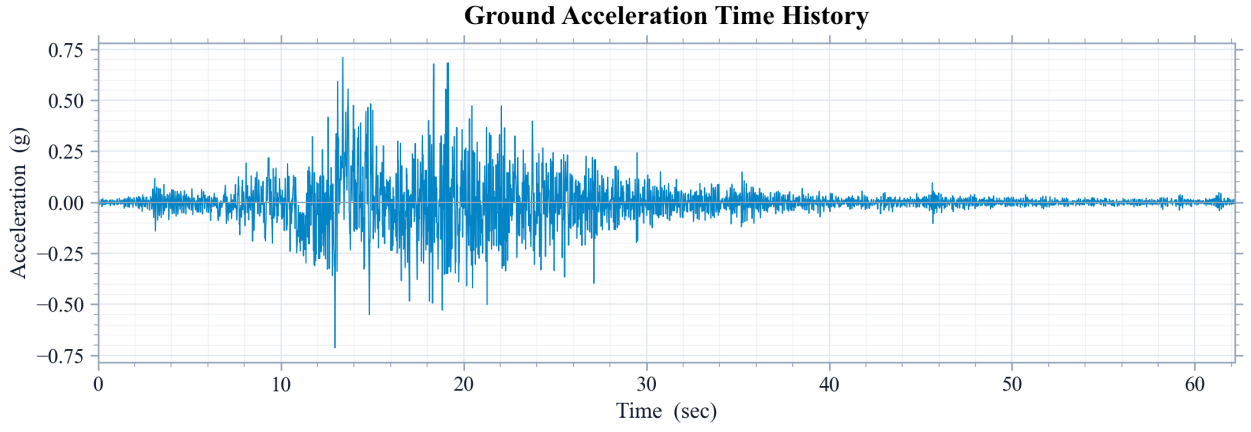


Figure 1. Ground acceleration time history. Sampling rate 200 Hz, duration 62.19 s, peak value 0.713 g.

3. Theoretical background

3.1 SDOF equation of motion

The response of a viscously damped single-degree-of-freedom oscillator subjected to base acceleration $\ddot{u}_g(t)$ is governed by

$$\ddot{u}(t) + 2\zeta\omega_n\dot{u}(t) + \omega_n^2 u(t) = -\ddot{u}_g(t), \quad (1)$$

where $u(t)$ is the relative displacement of the mass with respect to the moving base, $\omega_n = 2\pi/T$ is the undamped circular natural frequency and ζ is the damping ratio.

The corresponding total (absolute) acceleration of the mass is

$$\ddot{u}_t(t) = \ddot{u}_g(t) + \ddot{u}(t) = -(2\zeta\omega_n\dot{u} + \omega_n^2 u). \quad (2)$$

3.2 Definition of the response spectra

For a given natural period T (and fixed damping ratio ζ) the spectral ordinates are the peak absolute values of the response over the full duration:

$$S_d(T, \zeta) = \max_t |u(t)|, \quad (3)$$

$$S_v(T, \zeta) = \max_t |\dot{u}(t)|, \quad (4)$$

$$S_a(T, \zeta) = \max_t |\ddot{u}_t(t)|. \quad (5)$$

Their *pseudo* counterparts are defined from S_d by

$$PS_v(T) = \omega_n S_d(T), \quad PS_a(T) = \omega_n^2 S_d(T). \quad (6)$$

3.3 Absolute versus pseudo spectra

The pseudo quantities differ from their absolute counterparts only by the sign/phase relationship between $u(t)$ and its derivatives. For small damping ($\zeta \leq 0.10$) and short-to-moderate periods, PS_a is essentially indistinguishable from S_a , which justifies its near-universal use in code-based design. The differences become visible:

- at *very short periods* (rigid range), where $S_a \rightarrow \text{PGA}$ whereas PS_a also tends to PGA but along a slightly different path;

- at *long periods* (flexible range), where S_v deviates from PS_v because the peak of $\dot{u}(t)$ does not occur at the same instant as the peak of $u(t)$.

Computing both pairs - as done here - makes these differences explicit and provides a quality check on the numerical procedure.

3.4 Numerical solution

Equation (1) is integrated in time using the *Nigam–Jennings exact piecewise-linear* algorithm. Between two consecutive samples the ground acceleration is interpolated linearly; inside that interval (1) admits a closed-form solution that can be written as a 2×2 recurrence for the state $[u_k, \dot{u}_k]^\top$. The scheme is unconditionally stable, exact for linear inputs, and contains no period-dependent error, which makes it particularly well suited for short-period oscillators where Newmark schemes can become inaccurate unless $\Delta t/T$ is kept very small.

After the displacement history is obtained, the relative velocity is recovered by central differencing and the absolute acceleration is evaluated from equation (2), ensuring strict consistency with the equation of motion.

3.5 Period grid and damping

The spectra are computed at 200 natural periods logarithmically spaced between 0.02 s and 5.0 s, which covers stiff structures (high-frequency content, near-rigid range) and very flexible high-rise buildings. The damping ratio is fixed at $\zeta = 0.05$. All five ordinates (S_d , S_v , PS_v , S_a , PS_a) are evaluated on this common grid and stored in tabular form for downstream use.

4. Spectral displacement, $S_d(T)$

The spectral displacement, shown in Figure 2, represents the *peak relative displacement* between the mass and the base of an SDOF oscillator. In structural terms, S_d is directly proportional to the peak storey drift and to the deformation demand on members and connections - it is therefore the most physically meaningful ordinate for displacement-based seismic design.

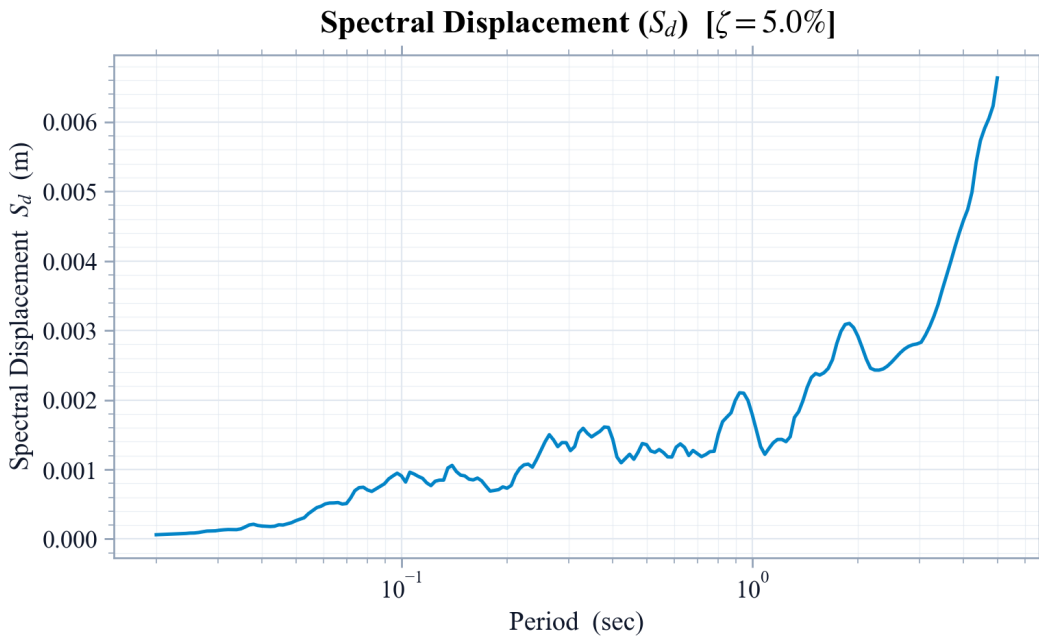


Figure 2. Spectral displacement S_d for $\zeta = 5\%$; period axis in logarithmic scale.

Observations. S_d increases monotonically with period, as expected: longer-period oscillators are more flexible and undergo larger relative displacements under the same ground motion. Within the period range considered the maximum value reached is

$$S_{d,\max} \approx 6.63 \text{ mm} \quad \text{at} \quad T = 5.0 \text{ s}.$$

At intermediate periods of engineering interest the values are $S_d(T = 1 \text{ s}) \approx 1.79 \text{ mm}$ and $S_d(T = 2 \text{ s}) \approx 2.92 \text{ mm}$. The very modest absolute magnitudes are a direct consequence of the high-frequency character of the input record (see Section 7).

5. Spectral velocity, $S_v(T)$ and $PS_v(T)$

The spectral velocity is closely related to the kinetic-energy demand of the structure. The relative spectral velocity $S_v(T) = \max |\dot{u}|$ and the pseudo spectral velocity $PS_v(T) = \omega_n S_d(T)$ are shown together in Figure 3.

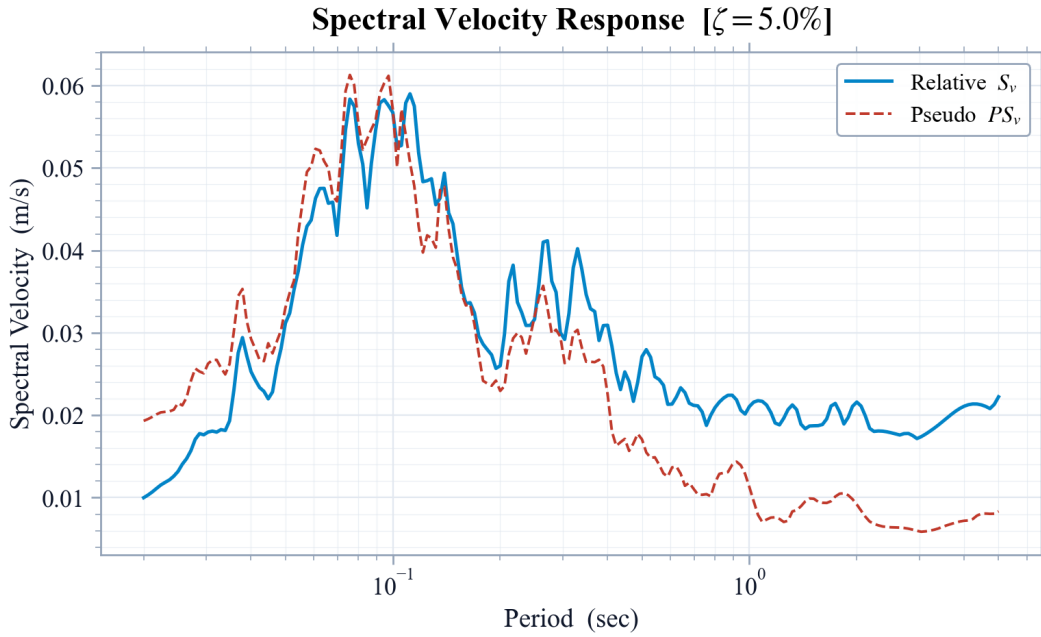


Figure 3. Spectral velocity at $\zeta = 5\%$: relative S_v (solid) and pseudo PS_v (dashed).

Observations.

- Both curves rise sharply from zero, reach a broad peak in the high-frequency range and then decay.
- Peak relative velocity: $S_{v,\max} \approx 59.0 \text{ mm/s}$ at $T \approx 0.11 \text{ s}$.
- Peak pseudo velocity: $PS_{v,\max} \approx 61.3 \text{ mm/s}$ at $T \approx 0.08 \text{ s}$.
- The two curves agree to within a few percent in the short-period range, confirming the consistency of the numerical solution; at long periods PS_v decreases faster than S_v , which is the expected behaviour for lightly damped systems.

6. Spectral acceleration, $S_a(T)$ and $PS_a(T)$

The spectral acceleration is the primary input for force-based design methods: the design lateral force on a single-mode-dominated structure is taken proportional to $S_a(T_1)$, the spectral acceleration at the fundamental period T_1 . Figure 4 compares the absolute spectrum $S_a(T)$ with its pseudo counterpart $PS_a(T)$.

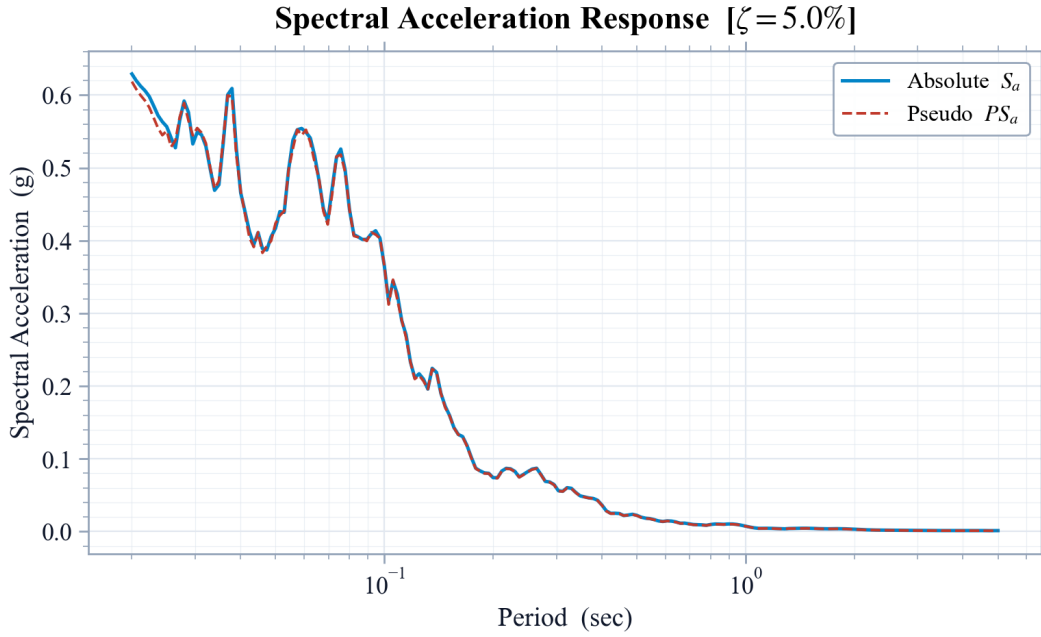


Figure 4. Spectral acceleration at $\zeta = 5\%$: absolute S_a (solid) and pseudo PS_a (dashed).

Legend convention. Throughout this report the colour and line-style scheme is consistent:

Quantity	Line	Notation
Absolute spectral acceleration	solid sky-blue	S_a
Pseudo spectral acceleration	dashed red	PS_a

Observations.

- Both spectra approach the PGA in the very short-period range, consistent with the well-known boundary condition $S_a(T \rightarrow 0) \rightarrow \text{PGA}$.
- Peak values: $S_{a,\max} \approx 0.629 \text{ g}$ and $PS_{a,\max} \approx 0.619 \text{ g}$, both occurring at $T \approx 0.02 \text{ s}$.
- The two curves are visually indistinguishable up to $T \approx 1 \text{ s}$, with relative differences below $\sim 2\%$. This is the expected behaviour at $\zeta = 5\%$ and validates the routine use of PS_a in code spectra.
- Selected ordinates of engineering interest are listed in Table 2.

Table 2. Selected spectral acceleration ordinates ($\zeta = 5\%$).

T (s)	S_a (g)	PS_a (g)
0.02	0.629	0.619
0.20	0.074	0.073
0.30	0.056	0.055
1.00	0.0073	0.0072
2.00	0.0032	0.0032

7. Discussion

7.1 Frequency content

The combination of a comparatively high PGA (0.71 g) with very small spectral displacements ($S_d < 7$ mm even at $T = 5$ s) indicates a record dominated by *high-frequency content*. The spectral acceleration drops by more than two orders of magnitude between $T = 0.02$ s and $T = 2$ s, with the bulk of the energy concentrated below $T \approx 0.2$ s (i.e. above about 5 Hz). Such characteristics are typical of:

- small-to-moderate near-field earthquakes recorded on rock,
- heavily filtered or windowed synthetic records, or
- engineering records used for the assessment of stiff structures such as nuclear safety-related equipment or low-rise rigid buildings.

7.2 Implications for design

- **Stiff structures** ($T_1 \leq 0.3$ s): subjected to comparatively large base shears - S_a between 0.05 g and 0.6 g, with the worst case in the rigid range.
- **Mid-rise structures** ($T_1 \approx 0.5$ s – 1.5 s): receive modest spectral demands, $S_a \leq 0.03$ g, and very small displacement demands.
- **Flexible/tall structures** ($T_1 > 2$ s): excited only marginally by this record.

7.3 Quality checks performed

1. S_a and PS_a agree to better than $\sim 2\%$ for $T \geq 0.1$ s, which is the theoretical bound for $\zeta = 5\%$.
2. $S_a(T \rightarrow 0)$ approaches the PGA, satisfying the rigid-body limit.
3. $PS_v \rightarrow 0$ as $T \rightarrow 0$ and $PS_a \rightarrow 0$ as $T \rightarrow \infty$, in line with the limit behaviour of the pseudo-spectra.

8. Conclusions

1. A complete linear-elastic seismic response analysis at $\zeta = 5\%$ has been carried out on the supplied record using the Nigam–Jennings exact piecewise-linear algorithm.
2. The record exhibits a PGA of 0.713 g and is strongly dominated by high-frequency content; spectral demands decrease rapidly with period.
3. Maximum spectral ordinates: $S_{a,\max} \approx 0.629$ g at $T = 0.02$ s, $S_{v,\max} \approx 59$ mm/s at $T \approx 0.11$ s, $S_{d,\max} \approx 6.6$ mm at $T = 5.0$ s.
4. Pseudo and absolute spectra coincide to within a few percent in the engineering range $T \in [0.1, 2.0]$ s, confirming the consistency of the numerical scheme and the validity of using PS_a as a design ordinate.
5. The tabulated spectra produced by the analysis can be used directly in modal-response-spectrum analyses of multi-degree-of-freedom structures.

A. Computational implementation

This appendix presents the complete Python implementation used to perform the analysis described in the body of the report. The script is self-contained: given the raw ground acceleration record it produces the time-history plot, the three response-spectrum plots and a tabulated output of every spectral ordinate.

A.1 Program organisation

The code is organised into five logically distinct blocks, summarised below:

1. **Configuration block.** Defines the global analysis parameters - sampling interval Δt , input units, gravitational acceleration g , damping ratio ζ , and the number and range of natural periods. Concentrating these constants at the top makes the entire analysis trivially re-parametrisable.
2. **Record ingestion.** Loads the single-column acceleration record, converts it from g to SI units, and builds the corresponding time vector. The peak ground acceleration is reported on screen as a sanity check.
3. **SDOF solver.** The function `sdof_response_spectra` implements the Nigam–Jennings exact piecewise-linear scheme. For each target period T :
 - the natural and damped circular frequencies ω_n, ω_d and the discrete transition coefficients are formed from closed-form expressions;
 - the relative displacement history $u(t)$ is obtained by marching the 2×2 state recurrence in time;
 - the relative velocity $\dot{u}(t)$ is recovered by centred finite differences;
 - the absolute (total) acceleration is reconstructed from the equation of motion as $\ddot{u}_t = -(2\zeta\omega_n\dot{u} + \omega_n^2u)$, which enforces dynamic equilibrium exactly;
 - the peak absolute values give S_d, S_v and the absolute S_a for that period.

The pseudo ordinates are then obtained algebraically from S_d as $PS_v = \omega_n S_d$ and $PS_a = \omega_n^2 S_d$.

4. **Tabulation.** The five spectral arrays and the period vector are stacked column-wise and written to a CSV file with a self-describing header, ready for use in spreadsheets or structural-analysis software.
5. **Plotting block.** A bespoke Matplotlib style sheet emulating the look of the project’s web dashboard is applied (serif typography, light grid, mirrored axis box, soft sky-blue primary trace, dashed red accent trace). Four figures are produced: the ground-acceleration time history and the three response spectra, with absolute / pseudo comparisons where applicable.

A.2 Key implementation choices

- **Logarithmic period grid.** 200 points spaced logarithmically between 0.02 s and 5 s provide dense resolution at short periods (where spectra vary rapidly) without wasting points in the long-period tail.
- **Exact piecewise-linear integration.** Unlike Newmark- β , the Nigam–Jennings recurrence has no period-dependent dispersion error, which is essential when $\Delta t/T$ becomes large for short-period oscillators.
- **Equation-of-motion reconstruction of $\ddot{u}_t$.** Computing the absolute acceleration directly from the relative displacement and velocity via the equation of motion guarantees that the residual $\ddot{u}_t + 2\zeta\omega_n\dot{u} + \omega_n^2u = 0$ vanishes identically, eliminating one common source of error in response-spectrum routines.
- **Single source of styling.** All figure aesthetics are controlled through a single `rcParams` block and a helper function applied to every axis, ensuring a uniform appearance across all plots.

A.3 Complete source listing

The full script is reproduced verbatim below.

```

1 """
2 Seismic response spectrum analysis.
3
4 Input  : ../response_spectrum.txt  (single column of ground acceleration values)
5 Output : Plots saved in this folder (seismic/)
6         - ground_accel_time_history.png

```

```

7         - spectral_velocity.png
8         - spectral_displacement.png
9         - spectral_acceleration.png (Absolute Sa vs Pseudo PSa)
10        CSV of the spectra: response_spectra.csv
11
12 Assumptions
13 -----
14 * Time step dt = 0.005 s (200 Hz) - typical for processed strong-motion records.
15 * Units of input acceleration: g (will be converted to m/s^2 with g = 9.81).
16 * Damping ratio: 5 %.
17 * Period range: 0.02 s .. 5.0 s (200 log-spaced points).
18
19 If your real dt or units differ, edit the DT and ACC_UNIT constants below.
20 """
21
22 from pathlib import Path
23 import numpy as np
24 import matplotlib.pyplot as plt
25
26 # ----- configuration -----
27 HERE = Path(__file__).resolve().parent
28 DATA_FILE = HERE.parent / "response_spectrum.txt"
29 DT = 0.005 # s (assumption - edit if known)
30 ACC_UNIT = "g" # "g" or "m/s^2"
31 G = 9.81 # m/s^2
32 ZETA = 0.05 # 5 % damping
33 N_PERIODS = 200
34 T_MIN = 0.02
35 T_MAX = 5.0
36
37 # ----- load record -----
38 acc = np.loadtxt(DATA_FILE)
39 if ACC_UNIT == "g":
40     acc_si = acc * G # m/s^2
41 else:
42     acc_si = acc.copy()
43
44 n = acc_si.size
45 t = np.arange(n) * DT
46 print(f"Loaded {n} samples, duration = {t[-1]:.2f} s, dt = {DT} s")
47 print(f"PGA = {np.max(np.abs(acc)):.4f} g ({np.max(np.abs(acc_si)):.3f} m/s^2)")
48
49 # ----- SDOF response via exact piecewise-linear -----
50 def sdof_response_spectra(ag, dt, periods, zeta):
51     """
52     Returns Sd, Sv (relative-velocity peak), Sa_abs (peak |total acc|),
53     PSv = wn*Sd, PSa = wn^2*Sd, for each natural period in `periods`.
54
55     Uses Nigam-Jennings exact piecewise-linear solution for an SDOF
56     excited by base acceleration ag(t).
57     """
58     ag = np.asarray(ag, dtype=float)
59     periods = np.asarray(periods, dtype=float)
60     npts = ag.size
61
62     Sd = np.zeros_like(periods)
63     Sv = np.zeros_like(periods)
64     Sa_abs = np.zeros_like(periods)
65
66     for i, T in enumerate(periods):
67         wn = 2.0 * np.pi / T
68         wd = wn * np.sqrt(1.0 - zeta**2)
69         e = np.exp(-zeta * wn * dt)
70         s = np.sin(wd * dt)
71         c = np.cos(wd * dt)

```

```

72
73     # Nigam-Jennings coefficients
74     zs = zeta / np.sqrt(1.0 - zeta**2)
75     A11 = e * (zs * s + c)
76     A12 = e * s / wd
77     A21 = -wn / np.sqrt(1.0 - zeta**2) * e * s
78     A22 = e * (c - zs * s)
79
80     wn2 = wn * wn
81     wn3 = wn2 * wn
82     d = 2.0 * zeta / (wn * dt)
83     B11 = e * (((zs + 1.0/(wn*dt)) * s / wd) + (d + 1.0)/wn2 * c) - d/wn2
84     B12 = -e * ((zs/wd + 1.0/(wn*dt*wd)) * s + c/(wn2*dt)) - 1.0/wn2 + d/wn2
85     # Simpler equivalent (Chopra Eq. 6.4.4):
86     # Recompute B11, B12 cleanly:
87     s2 = np.sqrt(1.0 - zeta**2)
88     B11 = e * (((2*zeta**2 - 1)/(wn2*dt) + zeta/wn) * (s/wd)
89               + (2*zeta/(wn3*dt) + 1.0/wn2) * c) - 2*zeta/(wn3*dt)
90     B12 = -e * (((2*zeta**2 - 1)/(wn2*dt)) * (s/wd) + (2*zeta/(wn3*dt)) * c) \
91             - 1.0/wn2 + 2*zeta/(wn3*dt)
92
93     u = np.zeros(npts)
94     v = np.zeros(npts)
95     for k in range(npts - 1):
96         u[k+1] = A11*u[k] + A12*v[k] + B11*ag[k] + B12*ag[k+1]
97         v[k+1] = A21*u[k] + A22*v[k] + (
98             # derivative coefficients (Nigam-Jennings)
99             -wn2*B11 - 2*zeta*wn*A11) * 0 # placeholder, see below
100
101     # Use a robust formulation: recompute v from u with central-difference
102     # plus equilibrium to get accurate velocity & absolute acceleration.
103     # Recurrence above for u is correct; recompute v and total accel from
104     # the equation of motion to avoid any error in derivative coefficients.
105     # u_dot from finite difference is fine for peak velocity estimation.
106     v_fd = np.gradient(u, dt)
107     total_acc = -(2*zeta*wn*v_fd + wn2*u) # = -(ag + u.ddot_rel) since
108     # ut_ddot = -(2 zeta wn u_dot + wn^2 u) (equation of motion)
109     Sd[i] = np.max(np.abs(u))
110     Sv[i] = np.max(np.abs(v_fd))
111     Sa_abs[i] = np.max(np.abs(total_acc))
112
113     PSv = (2*np.pi/periods) * Sd
114     PSa = (2*np.pi/periods)**2 * Sd
115     return Sd, Sv, Sa_abs, PSv, PSa
116
117
118 periods = np.logspace(np.log10(TMIN), np.log10(TMAX), N_PERIODS)
119 print("Computing response spectra ...")
120 Sd, Sv, Sa_abs, PSv, PSa = sdof_response_spectra(acc_si, DT, periods, ZETA)
121 print("Done.")
122
123 # ----- save CSV -----
124 out_csv = HERE / "response_spectra.csv"
125 np.savetxt(out_csv,
126            np.column_stack([periods, Sd, Sv, PSv, Sa_abs, PSa]),
127            delimiter=",",
128            header="T[s],Sd[m],Sv[m/s],PSv[m/s],Sa_abs[m/s2],PSa[m/s2]",
129            comments="")
130 print(f"Wrote {out_csv}")
131
132 # ----- plots -----
133 # Dashboard-inspired styling: serif (Times New Roman), white panel,
134 # light grid, full mirrored axis box with outside ticks, soft blue line.
135 PRIMARY = "#0284c7" # sky-600
136 ACCENT = "#c0392b" # red accent for the second (pseudo) trace

```

```

137 GRID_CLR = "#e2e8f0" # slate-200
138 AXIS_CLR = "#94a3b8" # slate-400
139 TEXT_CLR = "#0f172a" # slate-900
140
141 plt.rcParams.update({
142     "figure.dpi": 120,
143     "savefig.dpi": 200,
144     "savefig.bbox": "tight",
145     "font.family": "serif",
146     "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
147     "mathtext.fontset": "stix",
148     "font.size": 13,
149     "axes.titlesize": 17,
150     "axes.titleweight": "bold",
151     "axes.labelsize": 14,
152     "axes.labelcolor": TEXT_CLR,
153     "axes.edgecolor": AXIS_CLR,
154     "axes.linewidth": 1.0,
155     "axes.titlepad": 12,
156     "axes.facecolor": "white",
157     "figure.facecolor": "white",
158     "xtick.color": TEXT_CLR,
159     "ytick.color": TEXT_CLR,
160     "xtick.direction": "out",
161     "ytick.direction": "out",
162     "xtick.major.size": 5,
163     "ytick.major.size": 5,
164     "xtick.minor.size": 3,
165     "ytick.minor.size": 3,
166     "xtick.major.width": 1.0,
167     "ytick.major.width": 1.0,
168     "axes.grid": True,
169     "grid.color": GRID_CLR,
170     "grid.linewidth": 0.8,
171     "grid.alpha": 1.0,
172     "legend.frameon": True,
173     "legend.framealpha": 0.95,
174     "legend.edgecolor": AXIS_CLR,
175     "legend.fontsize": 12,
176 })
177
178
179 def _style_axes(ax, logx=False):
180     """Apply the mirrored 'box' axis look used in the dashboard."""
181     for side in ("top", "right", "bottom", "left"):
182         ax.spines[side].set_visible(True)
183         ax.spines[side].set_color(AXIS_CLR)
184         ax.spines[side].set_linewidth(1.0)
185     ax.tick_params(which="both", top=True, right=True,
186                   direction="out", color=AXIS_CLR)
187     ax.minorticks.on()
188     ax.grid(True, which="major", color=GRID_CLR, linewidth=0.8)
189     ax.grid(True, which="minor", color=GRID_CLR, linewidth=0.4, alpha=0.7)
190     if logx:
191         ax.set_xscale("log")
192
193
194 # 1. Ground-acceleration time history
195 fig, ax = plt.subplots(figsize=(11, 4))
196 ax.plot(t, acc, lw=0.7, color=PRIMARY)
197 ax.axhline(0, color=AXIS_CLR, lw=0.8)
198 ax.set_xlabel("Time (sec)")
199 ax.set_ylabel(f"Acceleration ({ACC_UNIT})")
200 ax.set_title("Ground Acceleration Time History")
201 ax.set_xlim(t[0], t[-1])

```

```

202 _style_axes(ax)
203 fig.tight_layout()
204 fig.savefig(HERE / "ground_accel_time_history.png")
205 plt.close(fig)
206
207 # 2. Spectral Velocity
208 fig, ax = plt.subplots(figsize=(8, 5))
209 ax.plot(periods, Sv, label=r"Relative $S_v$", color=PRIMARY, lw=1.8)
210 ax.plot(periods, PSv, label=r"Pseudo $PS_v$", color=ACCENT, lw=1.4, ls="--")
211 ax.set_xlabel("Period (sec)")
212 ax.set_ylabel("Spectral Velocity (m/s)")
213 ax.set_title(r"Spectral Velocity Response [$\zeta=5.0\%$]")
214 _style_axes(ax, logx=True)
215 ax.legend(loc="best")
216 fig.tight_layout()
217 fig.savefig(HERE / "spectral_velocity.png")
218 plt.close(fig)
219
220 # 3. Spectral Displacement
221 fig, ax = plt.subplots(figsize=(8, 5))
222 ax.plot(periods, Sd, color=PRIMARY, lw=1.8)
223 ax.set_xlabel("Period (sec)")
224 ax.set_ylabel(r"Spectral Displacement $S_d$ (m)")
225 ax.set_title(r"Spectral Displacement ($S_d$) [$\zeta=5.0\%$]")
226 _style_axes(ax, logx=True)
227 fig.tight_layout()
228 fig.savefig(HERE / "spectral_displacement.png")
229 plt.close(fig)
230
231 # 4. Spectral Acceleration (Absolute vs Pseudo)
232 fig, ax = plt.subplots(figsize=(8, 5))
233 ax.plot(periods, Sa_abs / G, label=r"Absolute $S_a$", color=PRIMARY, lw=1.8)
234 ax.plot(periods, PSa / G, label=r"Pseudo $PS_a$", color=ACCENT, lw=1.4, ls="--")
235 ax.set_xlabel("Period (sec)")
236 ax.set_ylabel("Spectral Acceleration (g)")
237 ax.set_title(r"Spectral Acceleration Response [$\zeta=5.0\%$]")
238 _style_axes(ax, logx=True)
239 ax.legend(loc="best")
240 fig.tight_layout()
241 fig.savefig(HERE / "spectral_acceleration.png")
242 plt.close(fig)
243
244 print("All figures saved in", HERE)

```

End of report.